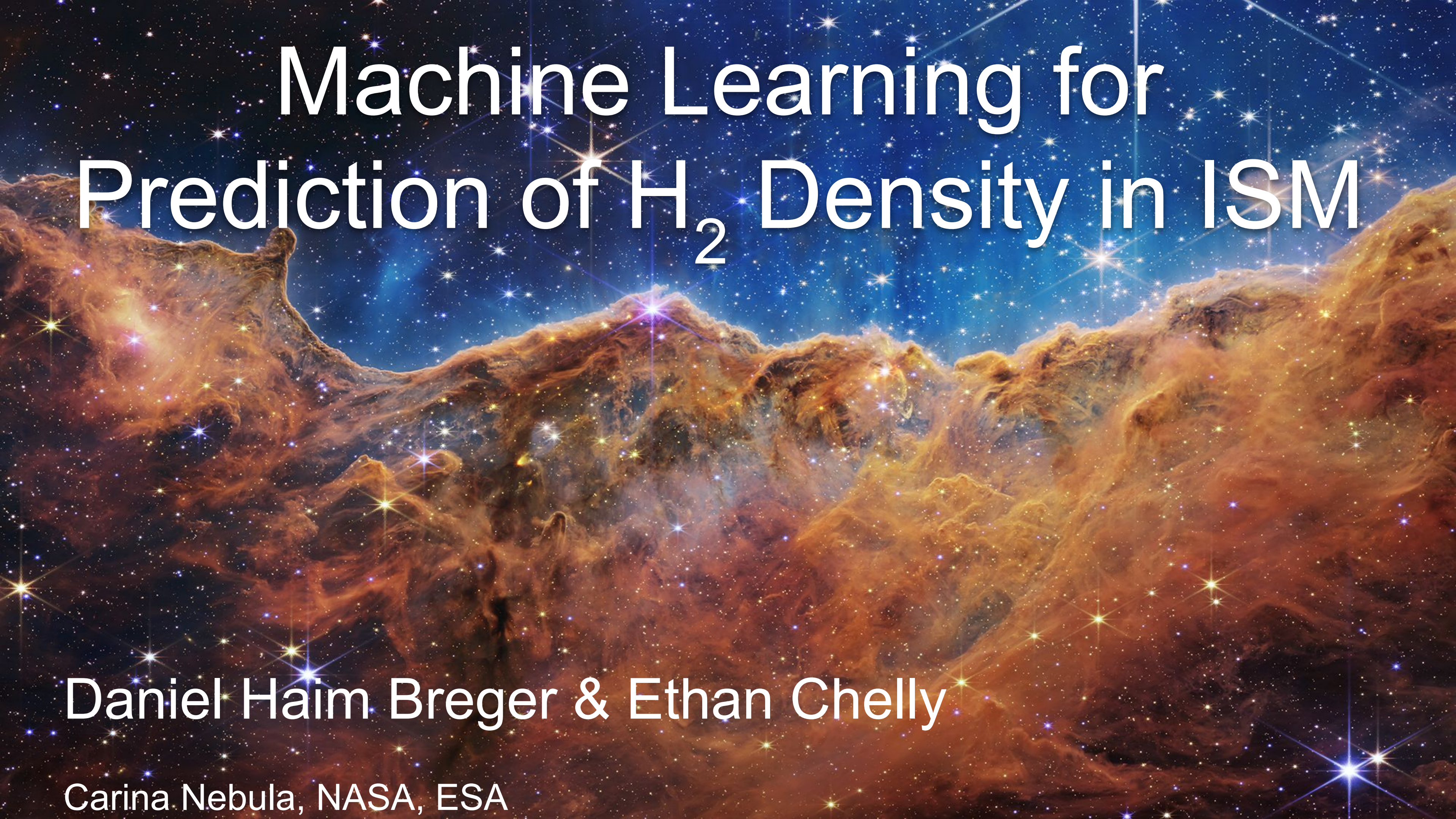
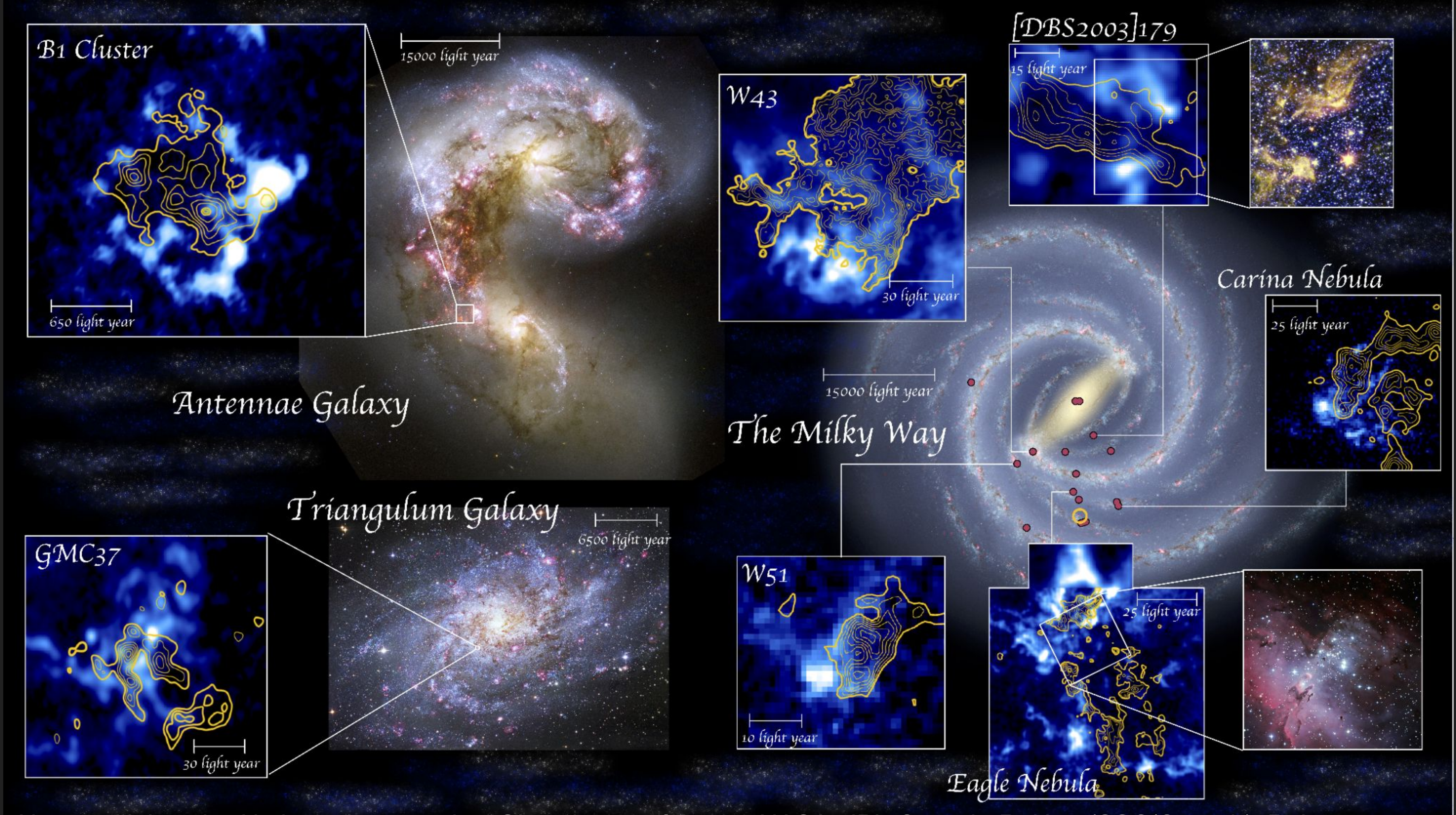
Machine Learning for Prediction of H_2 Density in ISM

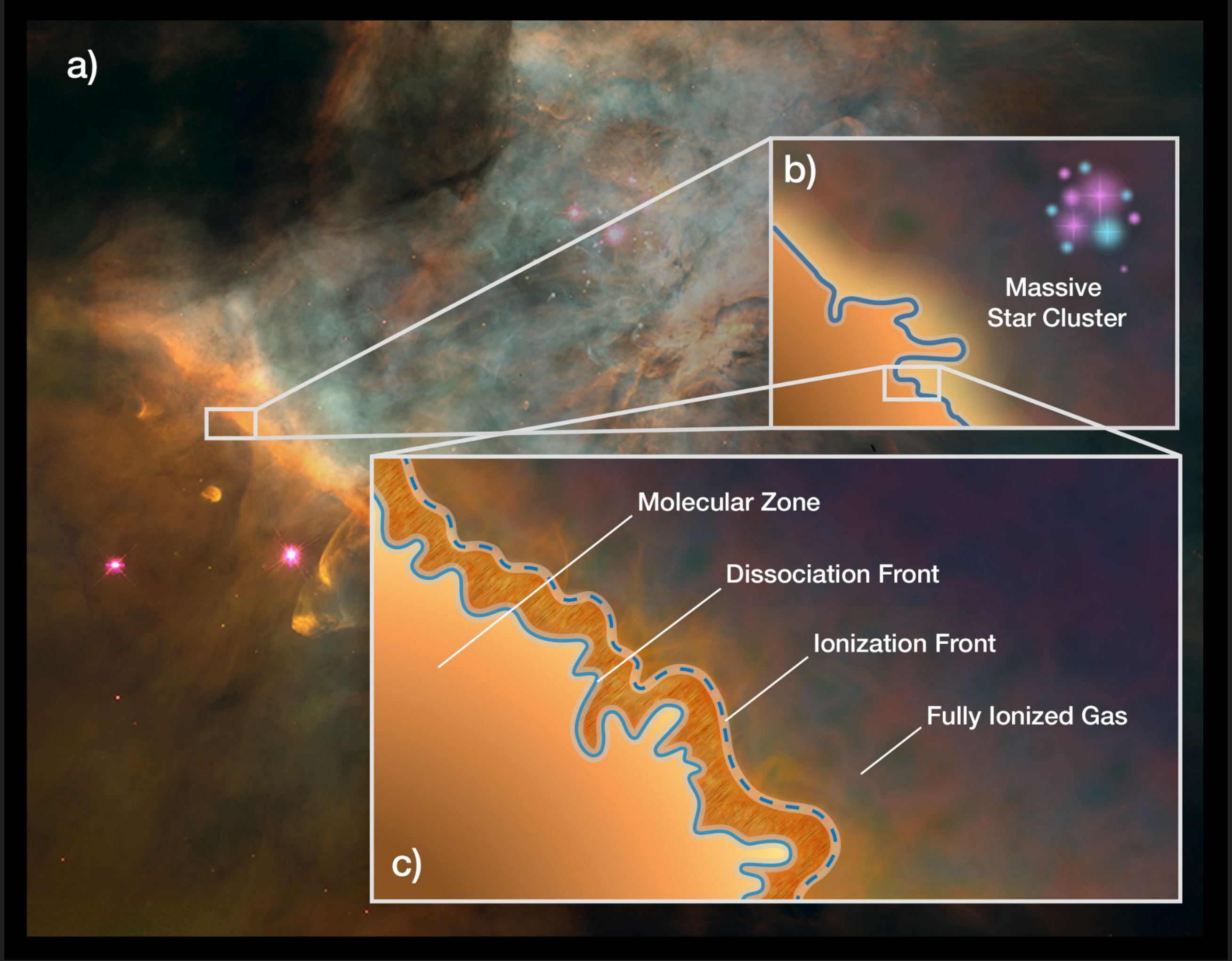
Daniel Haim Breger & Ethan Chelly



Carina Nebula, NASA, ESA

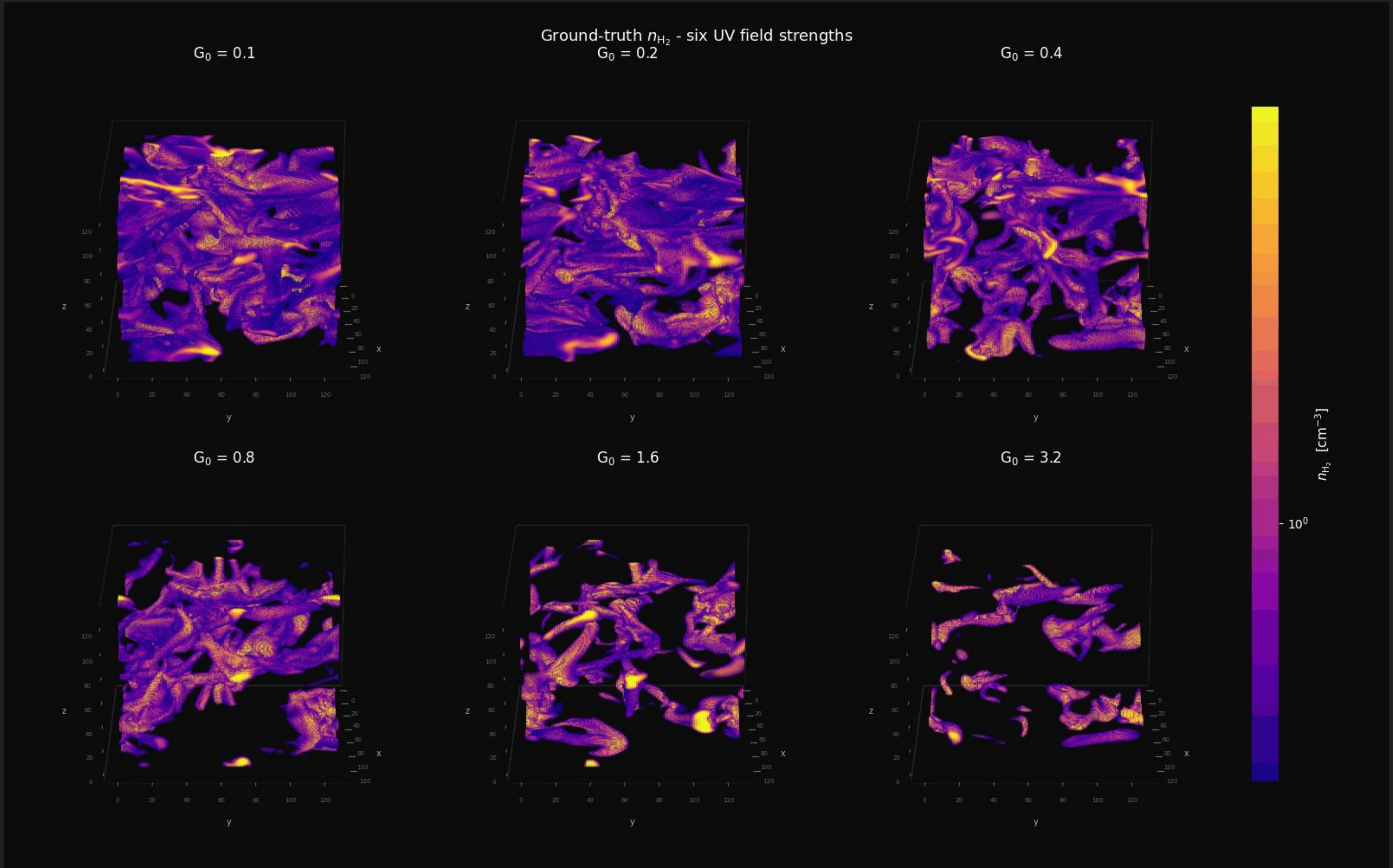


Nagoya University, National Astronomical Observatory of Japan, NASA, JPL-Caltech, R. Hurt (SSC/Caltech), Robert Gendler, Subaru Telescope, ESA, The Hubble Heritage Team (STScI/AURA), Hubble Collaboration, and 2MASS



NASA, ESA, CSA, PDRs4ALL ERS Team, Jason Champion (CNRS), Pam Jeffries (STScI)

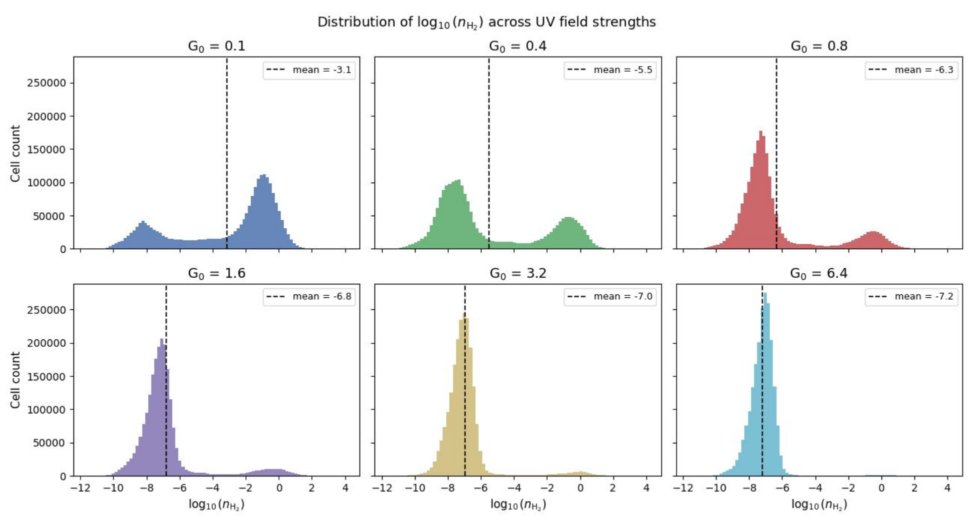
The Dataset



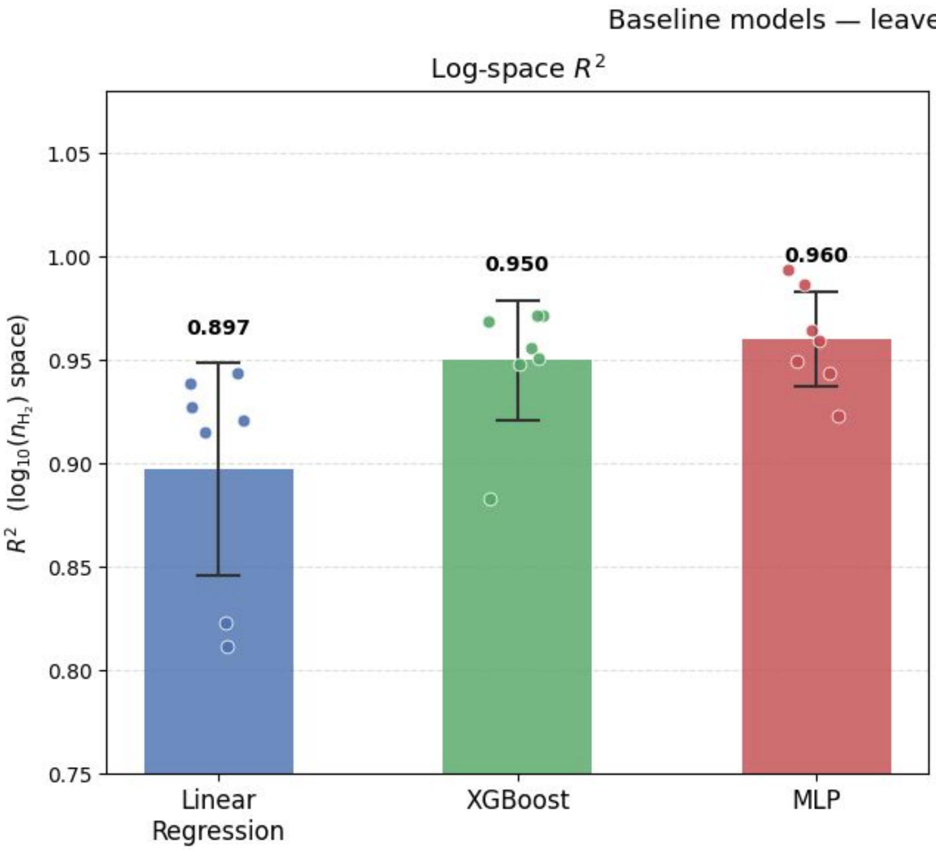
Ground-truth n_{H_2} - six UV field strengths
$G_0 = 0.1$
$G_0 = 0.2$
$G_0 = 0.4$
$G_0 = 0.8$
$G_0 = 1.6$
$G_0 = 3.2$
x
y
z
n_{H_2} [cm^{-3}]
10^0

Why This is Difficult

- Massive range of values
- Generalization for UV strength
- Spatial context matters

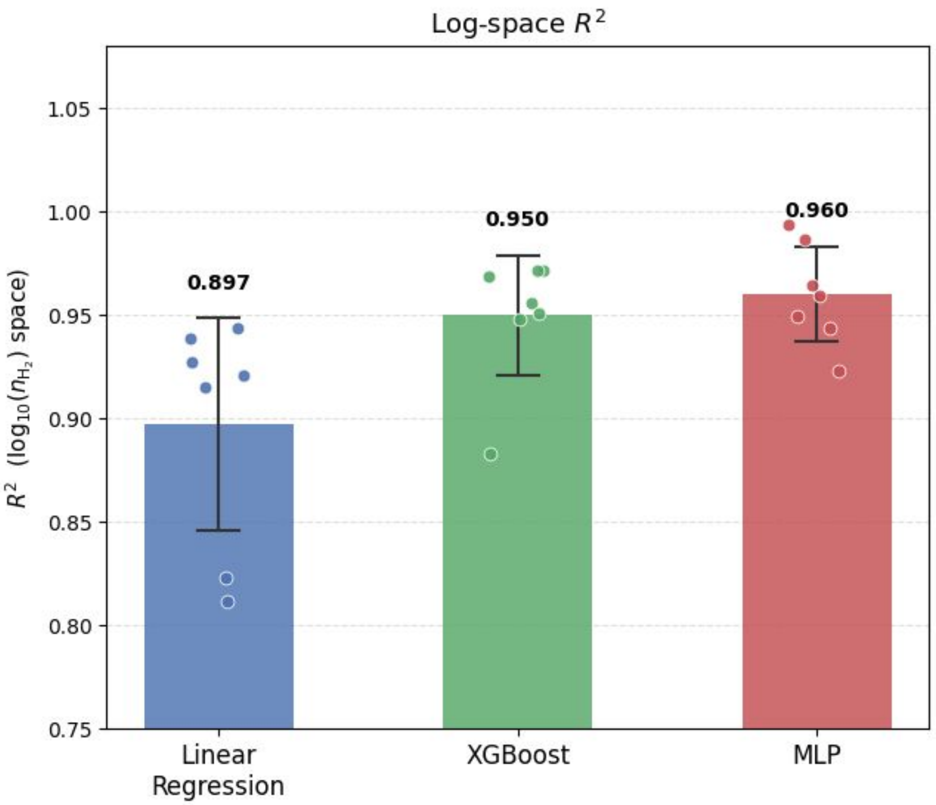


Baseline Models

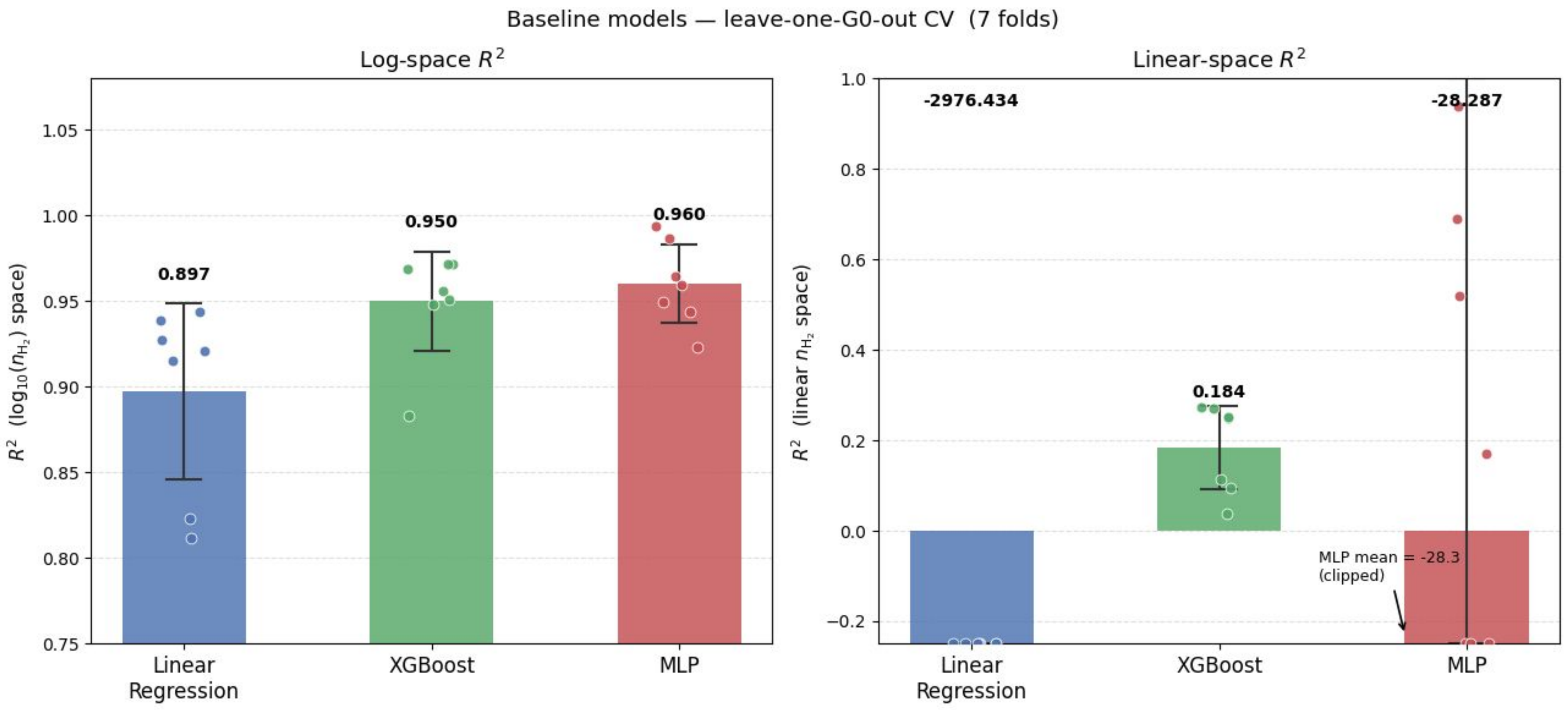


Baseline Models

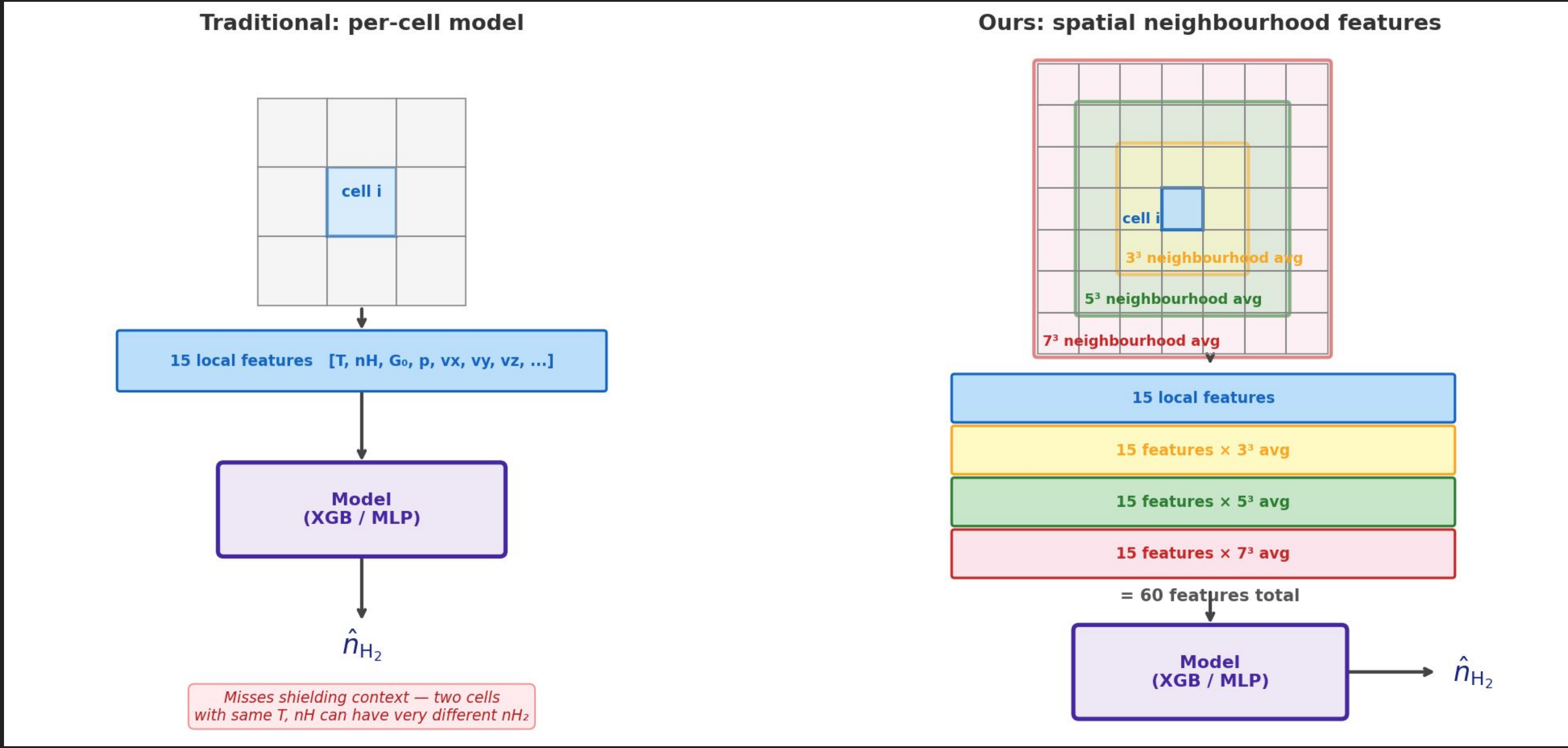
Baseline models — leave-one-G0-out CV (7 folds)



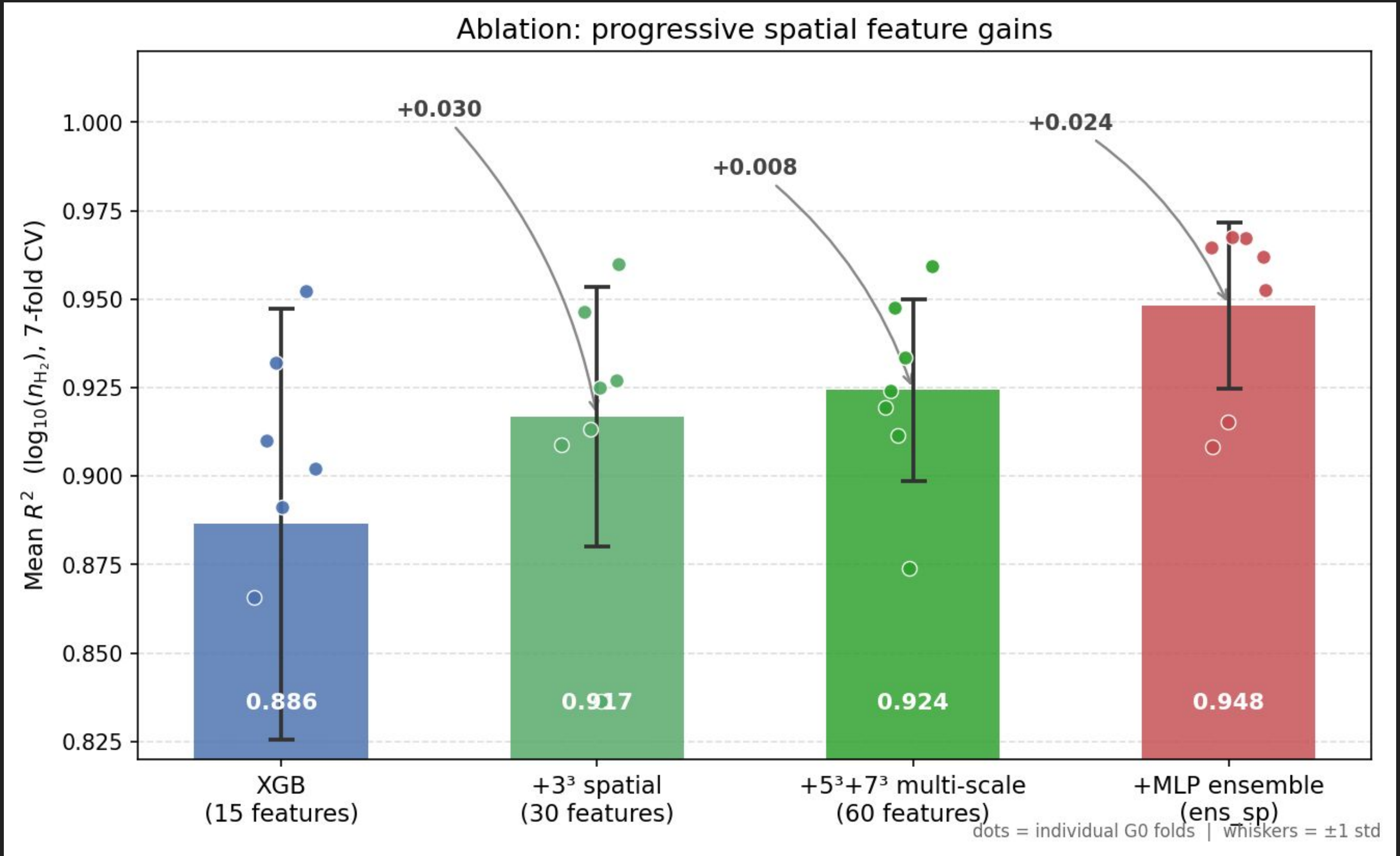
Baseline Models



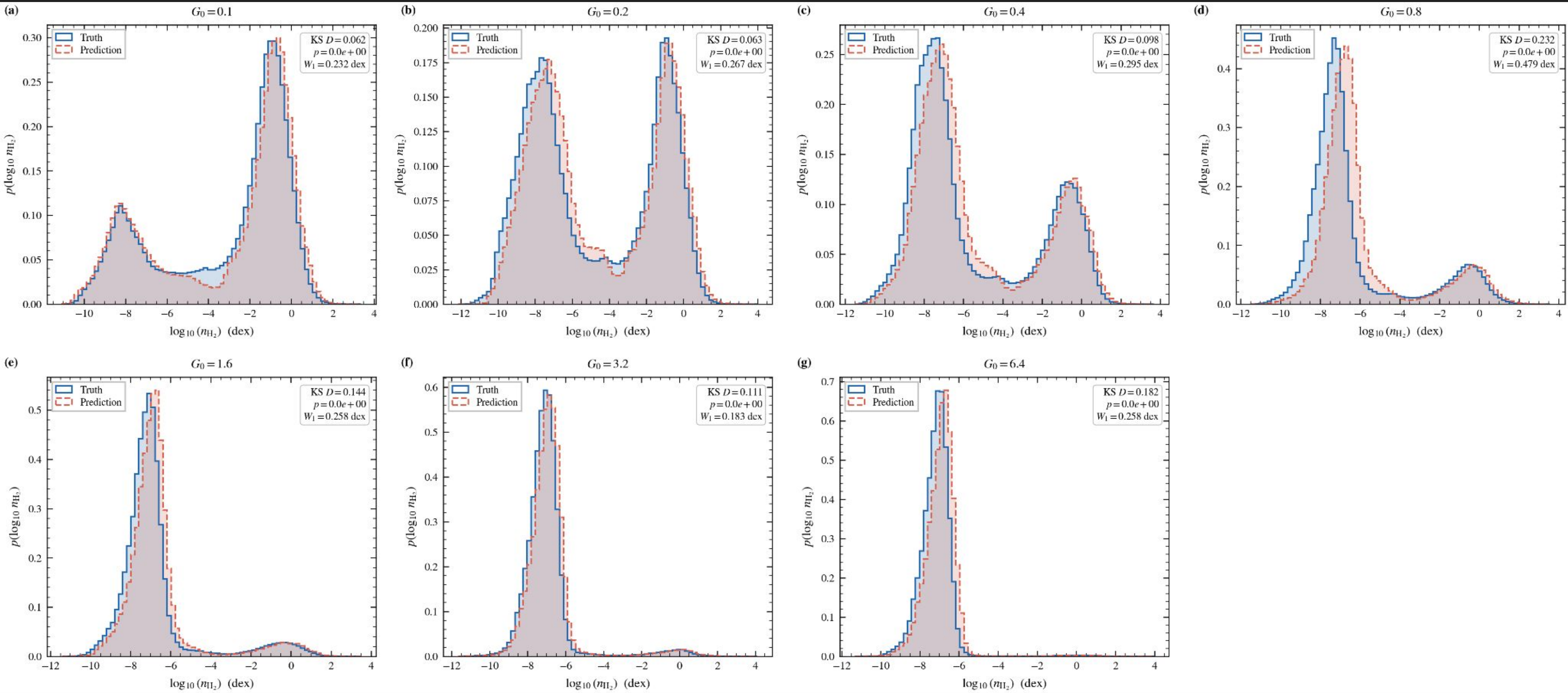
Spatial Features



Spatial Features

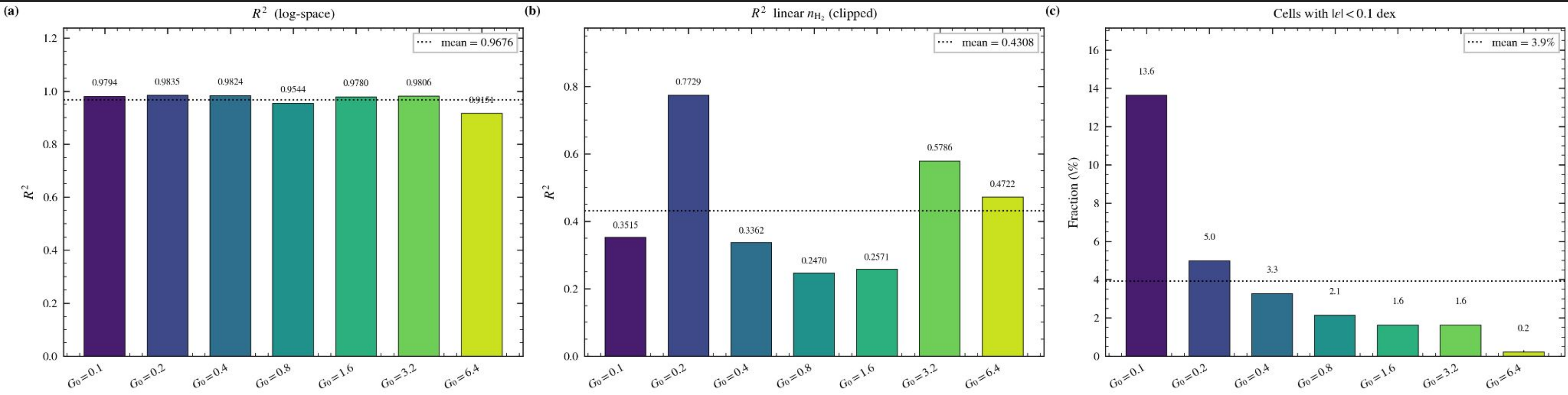


Results



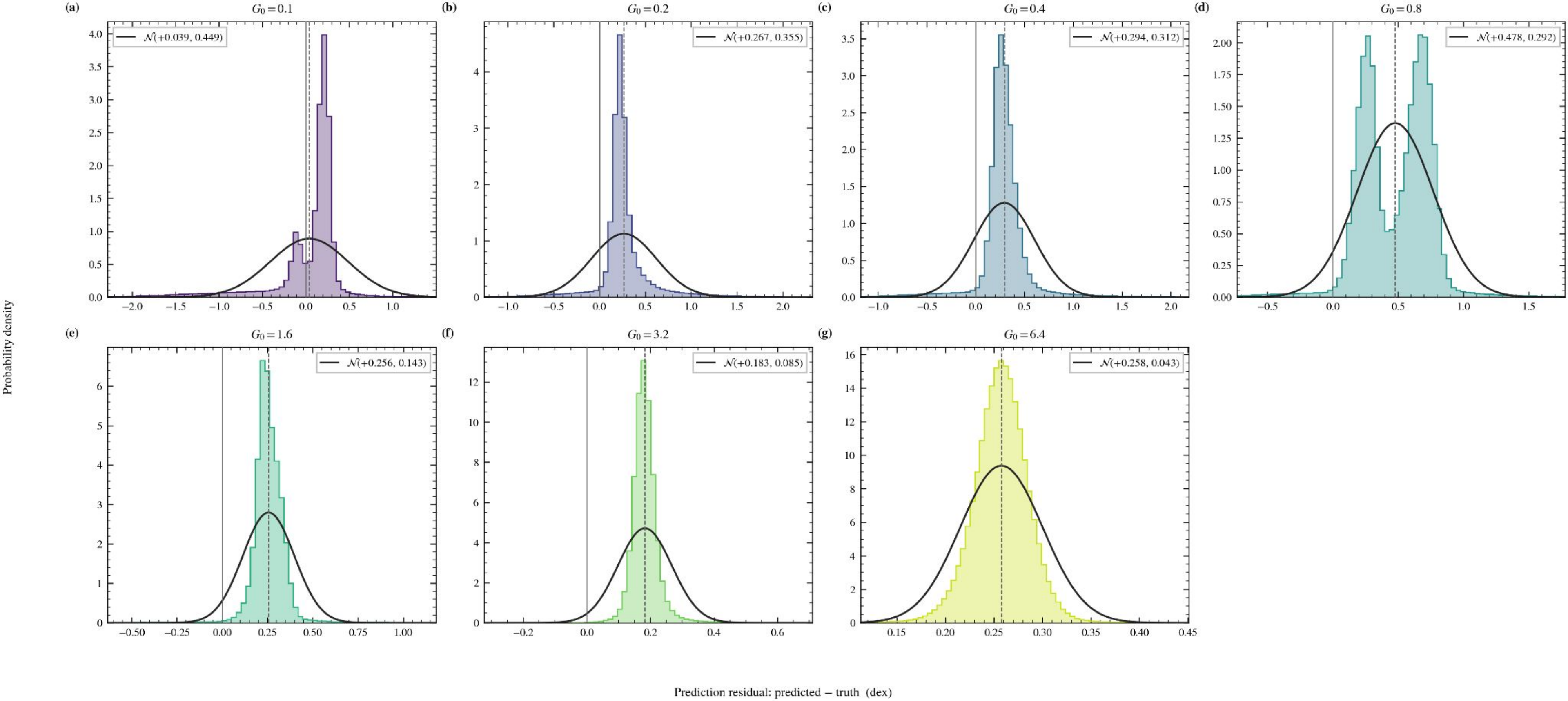
(a)
$G_0 = 0.1$
Truth
Prediction
KS $D = 0.062$
$p = 0.0e+00$
$W_1 = 0.232$ dex
(b)
$G_0 = 0.2$
KS $D = 0.063$
$p = 0.0e+00$
$W_1 = 0.267$ dex
(c)
$G_0 = 0.4$
KS $D = 0.098$
$p = 0.0e+00$
$W_1 = 0.295$ dex
(d)
$G_0 = 0.8$
KS $D = 0.232$
$p = 0.0e+00$
$W_1 = 0.479$ dex
(e)
$G_0 = 1.6$
KS $D = 0.144$
$p = 0.0e+00$
$W_1 = 0.258$ dex
(f)
$G_0 = 3.2$
KS $D = 0.111$
$p = 0.0e+00$
$W_1 = 0.183$ dex
(g)
$G_0 = 6.4$
KS $D = 0.182$
$p = 0.0e+00$
$W_1 = 0.258$ dex
$p(\log_{10} n_{H_2})$
$\log_{10}(n_{H_2})$ (dex)

Results

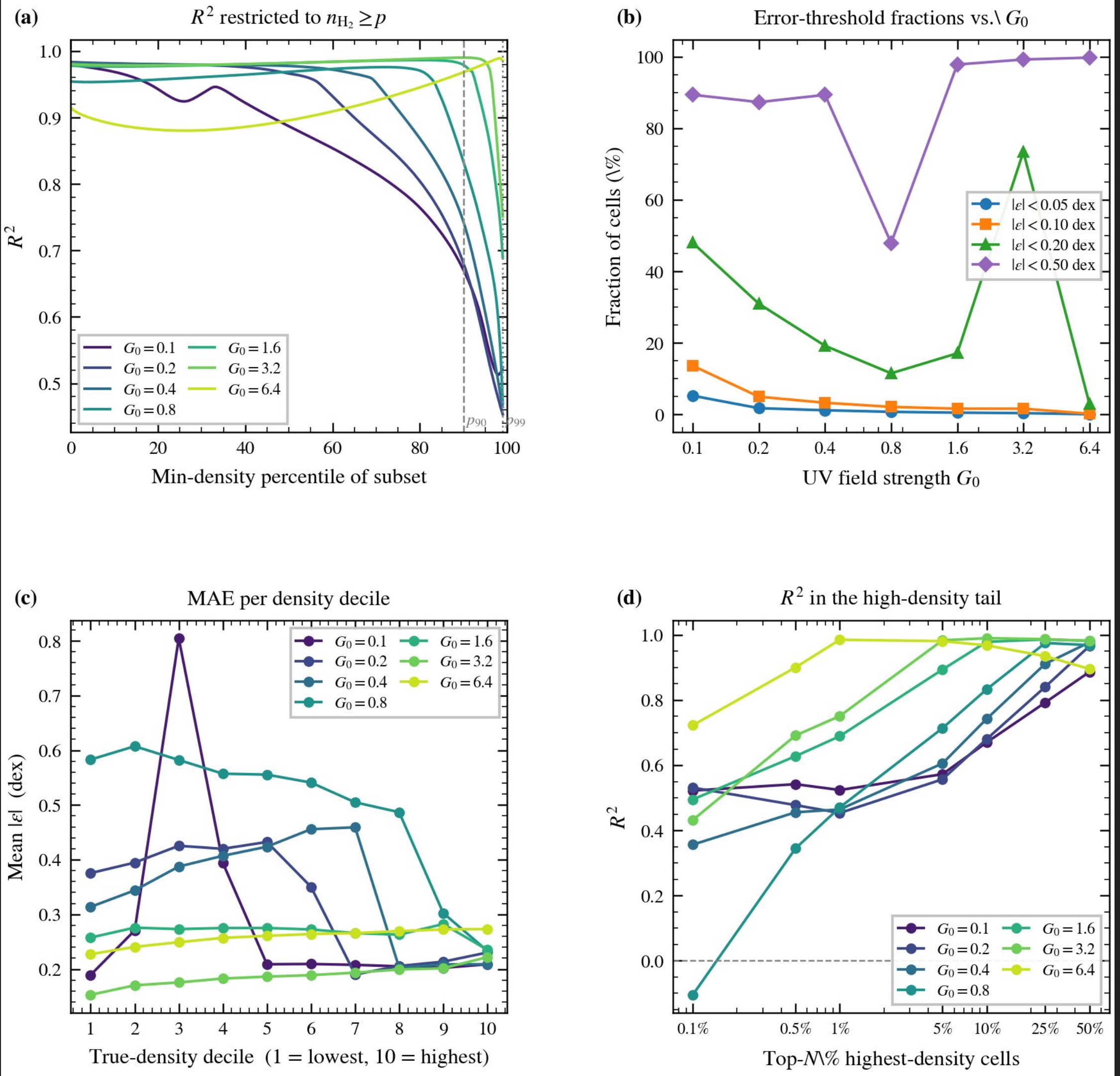


(a)
R² (log-space)
mean = 0.9676
R²
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0.9794
0.9835
0.9824
0.9544
0.9780
0.9806
0.9151
G₀=0.1
G₀=0.2
G₀=0.4
G₀=0.8
G₀=1.6
G₀=3.2
G₀=6.4
(b)
R² linear n_H₂ (clipped)
mean = 0.4308
R²
0.8
0.6
0.4
0.2
0.0
0.3515
0.7729
0.3362
0.2470
0.2571
0.5786
0.4722
G₀=0.1
G₀=0.2
G₀=0.4
G₀=0.8
G₀=1.6
G₀=3.2
G₀=6.4
(c)
Cells with |ε| < 0.1 dex
mean = 3.9%
Fraction (%)
16
14
12
10
8
6
4
2
0
13.6
5.0
3.3
2.1
1.6
1.6
0.2
G₀=0.1
G₀=0.2
G₀=0.4
G₀=0.8
G₀=1.6
G₀=3.2
G₀=6.4

Results

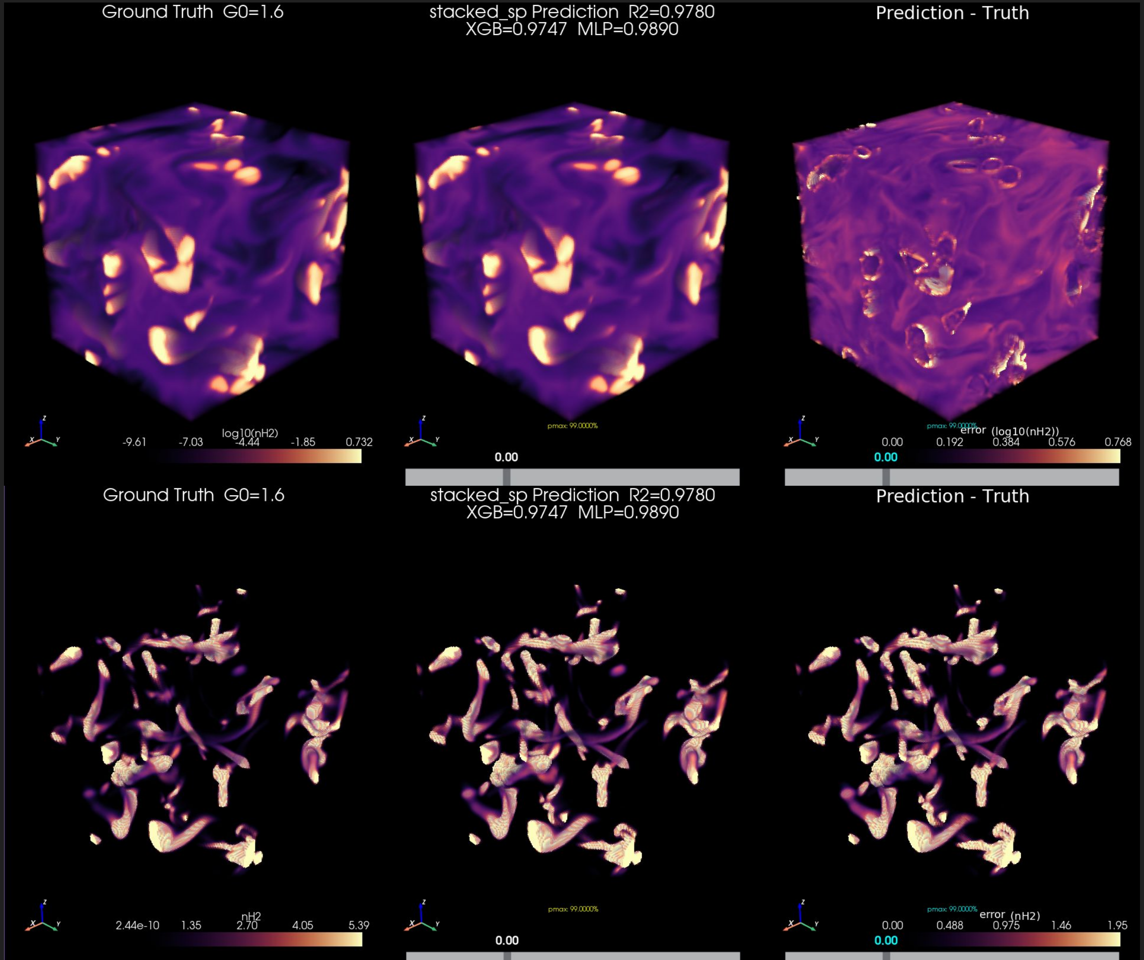

(a)
$G_0 = 0.1$
$\mathcal{N}(+0.039, 0.449)$
(b)
$G_0 = 0.2$
$\mathcal{N}(+0.267, 0.355)$
(c)
$G_0 = 0.4$
$\mathcal{N}(+0.294, 0.312)$
(d)
$G_0 = 0.8$
$\mathcal{N}(+0.478, 0.292)$
(e)
$G_0 = 1.6$
$\mathcal{N}(+0.256, 0.143)$
(f)
$G_0 = 3.2$
$\mathcal{N}(+0.183, 0.085)$
(g)
$G_0 = 6.4$
$\mathcal{N}(+0.258, 0.043)$
Probability density
Prediction residual: predicted − truth (dex)

Results



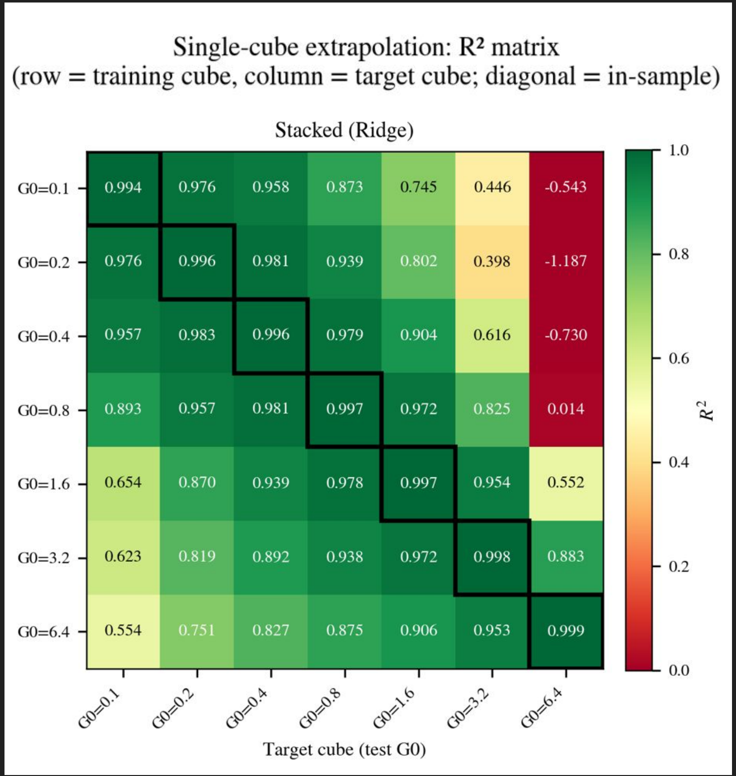
(a)
R^2 restricted to $n_{\mathrm{H}_2} \geq p$
R^2
Min-density percentile of subset
p_{90}
p_{99}
$G_0 = 0.1$
$G_0 = 0.2$
$G_0 = 0.4$
$G_0 = 0.8$
$G_0 = 1.6$
$G_0 = 3.2$
$G_0 = 6.4$
(b)
Error-threshold fractions vs.\ G_0
Fraction of cells (\%)
UV field strength G_0
$|\epsilon| < 0.05$ dex
$|\epsilon| < 0.10$ dex
$|\epsilon| < 0.20$ dex
$|\epsilon| < 0.50$ dex
(c)
MAE per density decile
Mean $|\epsilon|$ (dex)
True-density decile (1 = lowest, 10 = highest)
(d)
R^2 in the high-density tail
R^2
Top-N\% highest-density cells

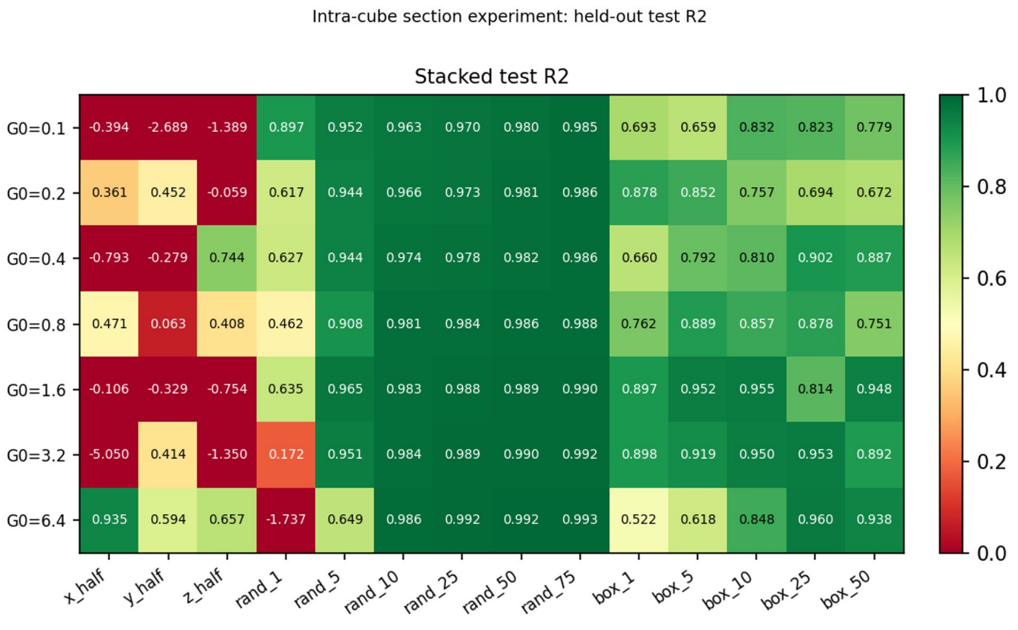
Results



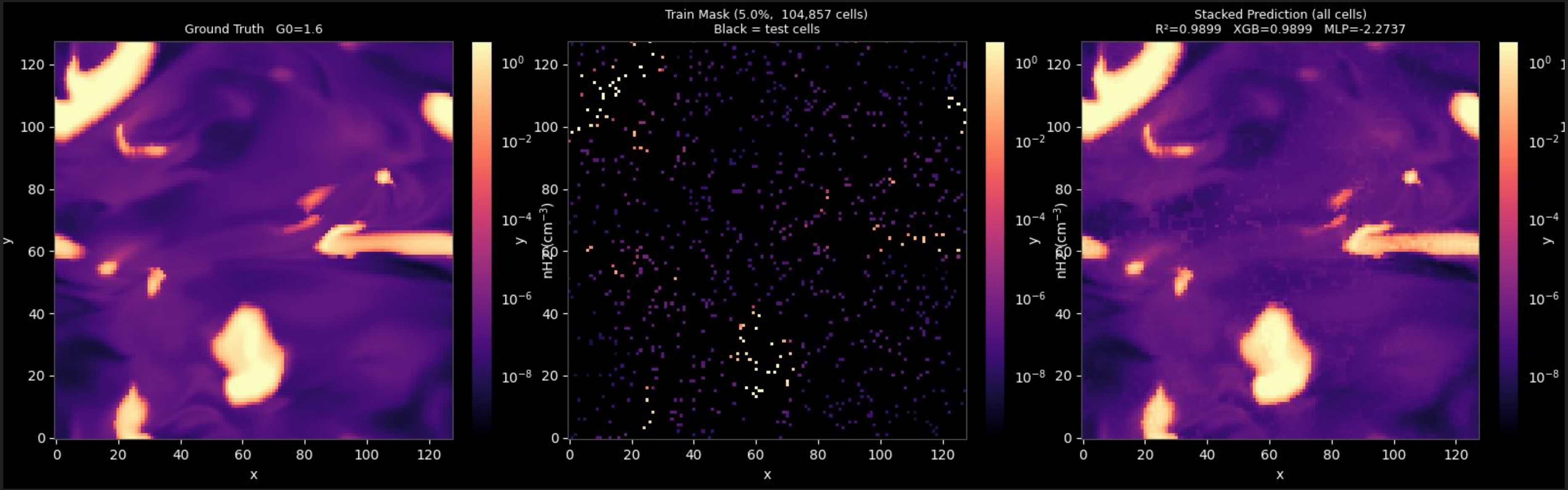
Ground Truth G0=1.6
stacked_sp Prediction R2=0.9780
XGB=0.9747 MLP=0.9890
Prediction - Truth
log10(nH2)
-9.61 -7.03 -4.44 -1.85 0.732
0.00
error (log10(nH2))
0.00 0.192 0.384 0.576 0.768
0.00
Ground Truth G0=1.6
stacked_sp Prediction R2=0.9780
XGB=0.9747 MLP=0.9890
Prediction - Truth
nH2
2.44e-10 1.35 2.70 4.05 5.39
0.00
error (nH2)
0.00 0.488 0.975 1.46 1.95
0.00

But Can We Extrapolate?

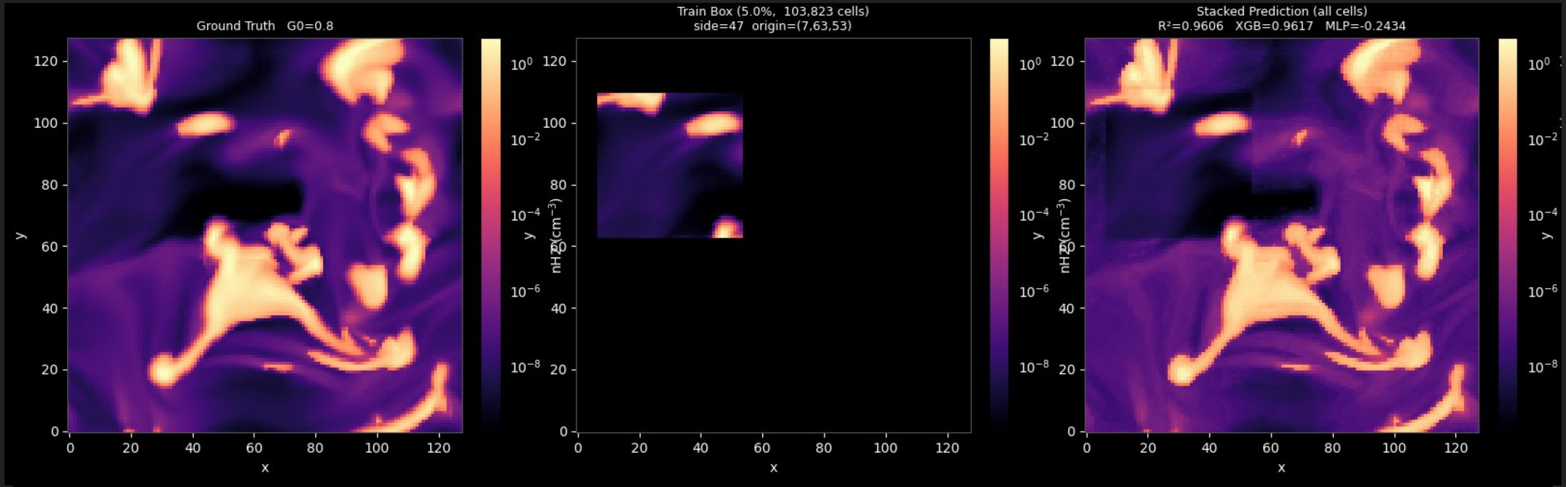


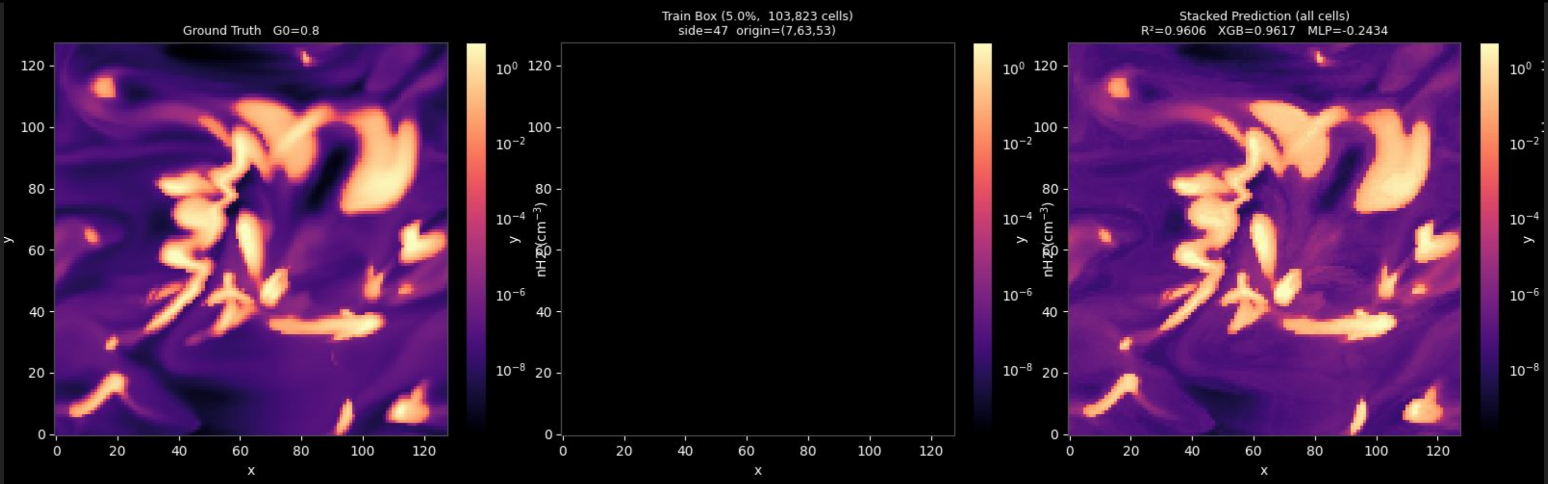


But Can We Extrapolate?

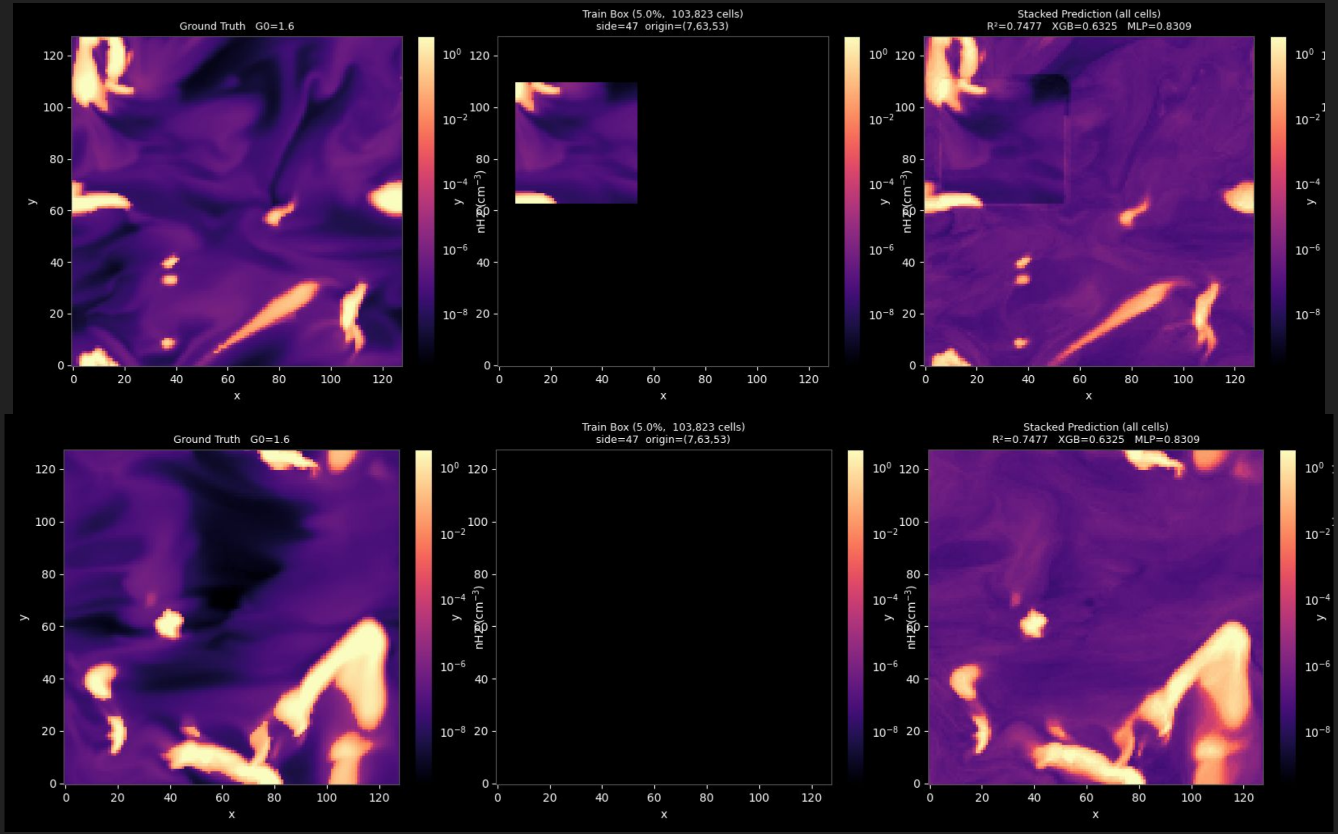


But Can We Extrapolate?

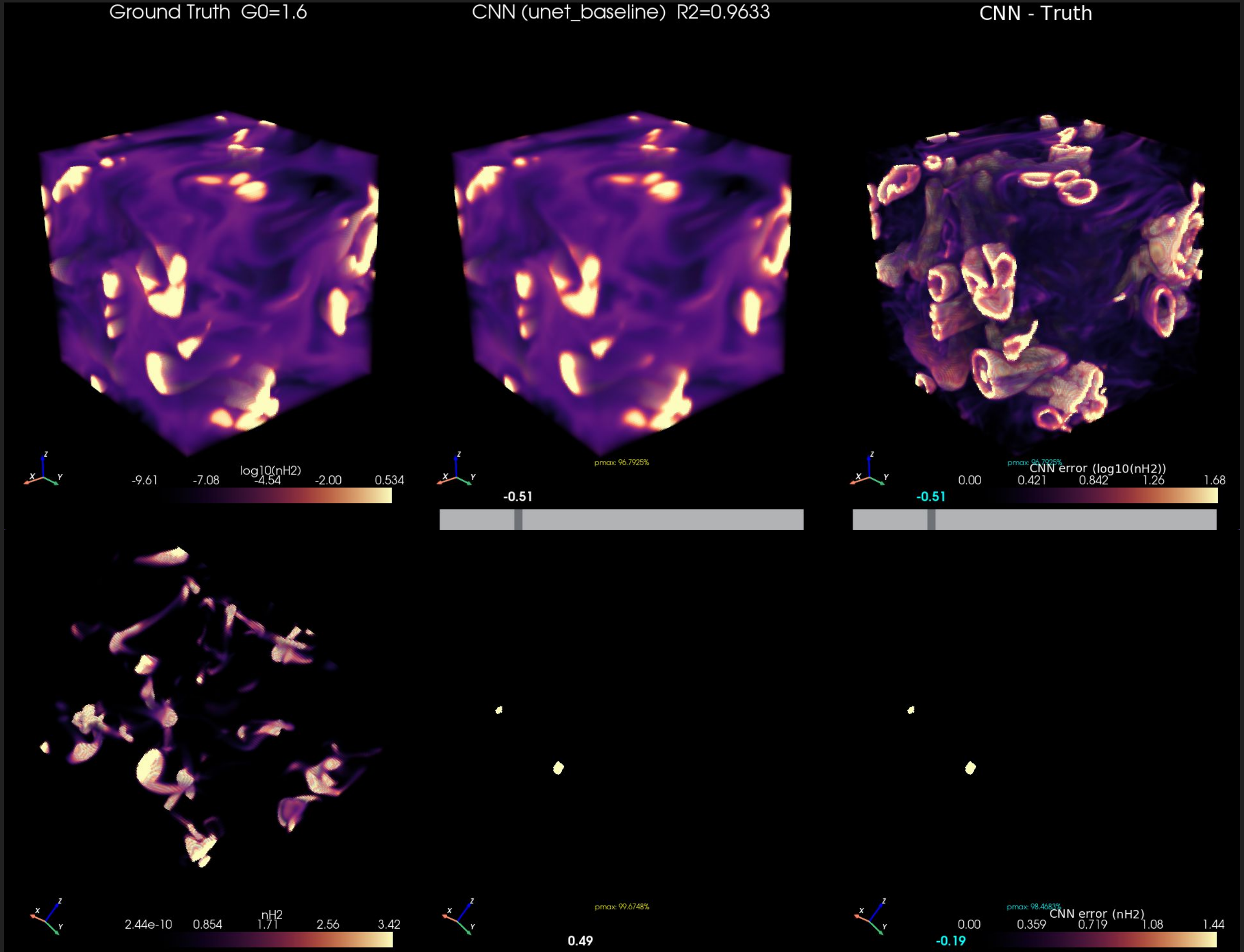




But Can We Extrapolate?



Why Not CNNs?



Ground Truth G0=1.6
CNN (unet_baseline) R2=0.9633
CNN - Truth
log10(nH2)
-9.61 -7.08 -4.54 -2.00 0.534
pmax: 96.7925%
-0.51
CNN error (log10(nH2))
0.00 0.421 0.842 1.26 1.68
-0.51
nH2
2.44e-10 0.854 1.71 2.56 3.42
pmax: 99.6748%
0.49
CNN error (nH2)
0.00 0.359 0.719 1.08 1.44
-0.19

Takeaways

- Yes, this is possible!
- Feature engineering > Architecture optimization
- Spatial context!
- CNNs good for structure, less for detail

Future Work

- Explore PINNs, FNOs, deeper ensembles
- Run with observational data
- Combine with MHD simulation pipelines
- Explore physics research implications

Thank You